our discussion). Ignoring this, both high-stake play and low-stake play seem reasonable. Are there animal gamblers who play for high stakes, and others with a more conservative game? In Chapter 9 we shall see that it is often possible to picture males as high-stake, high-risk gamblers, and females as safe investors, especially in polygamous species in which males compete for females. Naturalists who read this book may be able to think of species that can be described as high-stake, high-risk players, and other species that play a more conservative game. I now return to the more general theme of how genes make 'predictions' about the future.

One way for genes to solve the problem of making predictions in rather unpredictable environments is to build in a capacity for learning. Here the program may take the form of the following instructions to the survival machine: 'Here is a list of things defined as rewarding: sweet taste in the mouth, orgasm, mild temperature, smiling child. And here is a list of nasty things: various sorts of pain, nausea, empty stomach, screaming child. If you should happen to do something that is followed by one of the nasty things, don't do it again, but on the other hand repeat anything that is followed by one of the nice things.' The advantage of this sort of programming is that it greatly cuts down the number of detailed rules that have to be built into the original program; and it is also capable of coping with changes in the environment that could not have been predicted in detail. On the other hand, certain predictions have to be made still. In our example the genes are predicting that sweet taste in the mouth and orgasm are going to be 'good' in the sense that eating sugar and copulating are likely to be beneficial to gene survival. The possibilities of saccharine and masturbation are not anticipated according to this example; nor are the dangers of over-eating sugar in our environment where it exists in unnatural plenty.

Learning-strategies have been used in some chess-playing computer programs. These programs actually get better as they play against human opponents or against other computers. Although they are equipped with a repertoire of rules and tactics, they also have a small random tendency built into their decision procedure. They record past decisions, and whenever they win a game they slightly increase the weighting given to the tactics that preceded the victory, so that next time they are a little bit more likely to choose those same tactics again.